

Where the rain goes

Speaker notes — Al-Masar Al-Akhdar green corridor, Beirut

Built to be scanned, not read aloud. Each slide gives one **Say** line — the single message — then the figures to have ready, then the questions that slide reliably attracts. The *Key numbers* page at the end carries every number quoted on the slides.

SLIDE 1

Where the rain goes

Say: A physics-based flood model of the Fouad Boutros right-of-way, from a 39 GB LiDAR survey at half a metre. Three parts: how it works, how far it can be trusted, what it says about the corridor.

- 39 GB airborne LiDAR
- 0.5 m grid, 2.3 M cells

SLIDE 2

Chapter: how the simulator works

Say: Section break.

SLIDE 3

Rain falls on every cell

Say: Rain-on-grid: rainfall lands on every cell and the water finds its own way downhill. Nothing is pre-routed, so where flooding appears is a result, not an input.

- 9 land-cover classes set infiltration, roughness, detention
- 25 Nov 2025: 25.4 mm in 30 min - only a 2-5 year burst
- Mass balance closes to 3e-5 every run

Why does an ordinary storm flood the city?

Not extreme rainfall. Ordinary rain on steep, sealed streets with limited inlet capacity.

Rain on roofs?

Rerouted to the nearest street cell. Courtyards too.

SLIDE 4

What makes it different

Say: Two things. The physics other packages leave out - infiltration, detention, gullies that respond to head - and a solver you can differentiate, so it can design as well as predict.

- None of the 19 packages in the EA benchmark model infiltration or green infrastructure at all - they are hydraulics only
- None is differentiable; ours gives $d(\text{flooded area})/d(\text{design})$ to 2e-8
- Two schemes in one code, so scheme error is measured, not assumed
- Written in PyTorch on GPU - what makes the adjoint and ensembles affordable

Why does differentiability matter?

It turns the model from a predictor into a designer. Instead of testing a handful of layouts by hand you get the gradient of flooding with respect to every cell of the design, and can optimise directly. Nothing else in the field does this.

Isn't this just another SWE solver?

The hydraulics are standard and deliberately so - that is what lets us benchmark against everyone else. What is new is the surface model on top and the fact that the whole thing is differentiable.

What is it worse at?

No 1D sewer or river coupling, so two of the eight EA tests are out of scope. And the shock-capturing scheme costs about 3.5x the fast one.

SLIDE 5

The corridor, surface by surface

Say: The design is written into the terrain cell by cell - seven surface types, each with its own infiltration, roughness and detention depth.

- 31,828 m² total
- Rain garden is 52% of it
- Bioswales + ponds together under 4,000 m²

How faithful is the layout?

SAY THIS UNPROMPTED. Zone polygon and alignment are real data. The cross-section - band widths, where swales sit, pond and terrace placement - is our reading of the published drawings plus what seemed hydraulically sensible. Faithful in spirit, not a construction spec. If the design team shares real dimensions it is a data swap and a re-run, not a rebuild.

Which element does the work?

Mostly plain permeable ground, not the engineered features.

SLIDE 6

Chapter: validation

Say: Section break.

SLIDE 7

Against an independent solver

Say: The strongest evidence here. A separate code, written by a different group, solving the full equations with a different scheme, on the identical grid and storm.

- HLLC vs SynxFlow: IoU 0.913, corr 0.985, RMSE 3.7 cm
- Inertial vs SynxFlow: IoU 0.690, RMSE 7.8 cm, +27% wet area
- LISFLOOD-FP: fabricated 6x the input volume, failed its own mass balance

Why is the fast scheme biased?

It drops the advective momentum term, so it smears the flood front. Matters here because these streets run supercritical at 6-8 m/s on a 6% grade.

Is LISFLOOD's failure our fault?

No - documented limitation of its scheme on terrain with retaining walls. Its raster is void rather

than merely different.

SLIDE 8

Benchmark: Test 8A

Say: The EA case closest to what we do: direct rainfall on a real dense street network. We are inside the main cluster at all four gauges with published curves.

- P1 0.579 · P2 0.239 · P3 0.726 · P6 0.064 m
- 19 packages in the published comparison
- Dark bar excludes the report's own approximate codes

Why only four points?

Only four have published time series in the report.

How accurate are the bands?

Read off the report's figures, so good to a centimetre or two. We sit well inside them, not on the edges.

SLIDE 9

Benchmark: Test 4

Say: Tests how fast a flood front travels and how deep the water is behind it. In cluster at all nine points, on both schemes.

- Built from the written spec - the EA supplies no terrain file for this case
- 5 of 5 analytic tests pass: conservation, Manning normal depth, well-balancedness, reference-port equivalence

Why not all eight EA tests?

The rest need data files we do not have, and two need coupled 1D sewer or river models we do not implement. Test 3 is the one worth acquiring - it discriminates shock-capturing schemes, which is exactly our new capability.

SLIDE 10

Resolution

Say: The EA report says it is unclear whether finer than 2 m is worth it, and recommends some-one test at 0.5 m. This is that test.

- 0.5 m: 9.4% benefit · 2 m: 0.6%
- Peak velocity climbs 4.3 -> 14.8 m/s as the grid coarsens
- Same terrain coarsened, so only the grid changes

Why does velocity go up?

Block-averaging manufactures artificial steep gradients while erasing the kerbs and narrow streets that convey the water. Wrong direction and physically implausible - that is the point.

Is this a fair test?

Caveat honestly: coarsening is not identical to building natively at 2 m, where roughness would be recalibrated. Direction and size are unambiguous.

SLIDE 11

Chapter: Beirut

Say: Section break.

SLIDE 12

Before and after

Say: The observed storm, before and after, difference on the right. Blue is drier.

- Effect extends beyond the ribbon into surrounding streets
- The corridor follows a natural drainage axis - it is already where the water wants to go

What is the large square top-left?

A real deep excavation on a construction site, surveyed by the LiDAR and modelled as a basin. Genuine terrain, not an artefact.

SLIDE 13

Results

Say: Over half the flooding removed underfoot in a frequent storm, about half in the observed event, essentially nothing in a 50-year storm - which is the next slide.

- Benefit reaches roughly two blocks, then fades
- Metric excludes the fifth of the corridor dug out to hold water
- Those cells are 20% of the ribbon but 34% of its wet area

Why exclude the ponds?

Counting a full bioretention pond as flooding penalises the design for working. The published study drew the same distinction.

SLIDE 14

The severe storm, and a correction

Say: In a 50-year storm the corridor stops keeping itself dry and starts protecting everywhere else. Two different jobs; only the second survives an extreme event.

- T50: area barely moves, but +2,124 m3 absorbed and outflow cut 11%
- Infiltration multiple is 1.3x, NOT the 3x previously claimed
- Absolute gain (~900 m3 observed storm) holds up

Why did the infiltration claim shrink?

Better soil physics. Ordinary Beirut ground absorbs far more than a constant rate implied, so the corridor's relative contribution is smaller. Better physics, smaller claim - say it plainly.

SLIDE 15

What would make this a validated model --- the ask

Say: Verified, not validated. It solves the equations correctly and agrees with an independent solver to 3.7 cm - but every comparison is against another model, never a measured flood in

Beirut.

- Needed: water marks at a few dozen streets after the next big storm
- Plus a handful of low-cost depth loggers, and timing
- No specialist equipment, no large team

Can we use the numbers meanwhile?

Yes, with care. The study reports differences, and a systematic bias affects both scenarios and largely cancels. Absolutes carry more uncertainty.

How can we help?

OFFER THIS. The model already predicts where flooding concentrates, so we can site the loggers and surveys efficiently. Makes it a collaboration, not a favour.

SLIDE 16

Close

Say: Thank you / questions.

SLIDE REF

Key numbers

Corridor results — flooded-area reduction, best-validated configuration (shock-capturing scheme with Green–Ampt infiltration). “Walkable” excludes the fifth of the corridor dug out as swales and ponds, which hold water by design.

Storm	Walkable	Whole ribbon	Extra absorbed	Outflow cut
frequent (T2)	58.1%	38.6%	+911 m ³	18%
observed 25 Nov 2025	52.0%	29.5%	+919 m ³	17%
severe (T50)	-0.7%	-13.5%	+2124 m ³	11%

Validation — against SynxFlow, an independently written full shallow-water solver, on the identical 0.5 m grid and storm.

Scheme	Extent IoU	Correlation	RMSE	Wet-area bias
Fast inertial (3-term)	0.690	0.938	7.8 cm	+27%
Shock-capturing (HLLC)	0.913	0.985	3.7 cm	+1%

Grid resolution — same terrain, coarsened; only the computational grid varies.

Resolution	Cells	Corridor benefit	p99 street velocity
0.5 m	2 285 142	9.4%	4.3 m/s
1.0 m	571 000	2.4%	8.1 m/s
2.0 m	142 500	0.6%	11.9 m/s
4.0 m	35 500	2.4%	14.8 m/s

If you are asked something not covered here, the safe answer is that the model is verified against analytic solutions and agrees closely with an independent solver, but has never been tested against a measured flood in Beirut, because no such measurement exists. Say that rather than improvise.